

AWS IAM n8n Automation Prototype - Full Project Overview

A portfolio-grade multi-workflow SOC automation prototype covering triage, containment, hygiene monitoring, and TheHive case-closure feedback

Project done by: [Abdul Rehman \(abdul4rehman215\)](#)

GitHub: [abdul4rehman215](#)

AWS IAM n8n Automation Project Overview

Document Type

Master prototype document | Full project overview

Prototype Focus

AWS IAM automation | n8n | TheHive 5 | Slack

Project Character

Portfolio prototype with implementation, testing, and validation evidence

Design Goal

Showcase an end-to-end SOC automation workflow with clear business and analyst value

Why this document matters

- This write-up is designed to work both as a project portfolio document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the operational value to a SOC, and the validation evidence that proves the workflow worked in the lab.
- The screenshots included throughout the document are the actual workflow, node, Slack, DataTable, AWS, and TheHive validation artifacts collected during the build process.

Key outcomes

- Combines four connected workflows into a single prototype operating model for cloud identity security operations.
- Uses n8n as the orchestration layer between AWS, Slack, DataTable, and TheHive 5.
- Demonstrates detection, triage, response, preventive hygiene monitoring, and case-closure synchronization in one project.

1. Project Summary

This master document describes the complete AWS IAM automation prototype built across four connected n8n workflows. The design was approached as a practical SOC portfolio project, but the resulting prototype also demonstrates a realistic operating model for cloud identity security operations.

Instead of presenting four isolated automations, this document explains how they fit together: Flow A handles event-driven triage, Flow B performs containment and case promotion, Flow C performs recurring preventive IAM hygiene monitoring, and Flow D synchronizes case-closure outcomes back into the shared operations state.

Taken together, the prototype shows a closed-loop workflow lifecycle: detect -> enrich -> score -> notify -> track -> contain -> escalate -> monitor hygiene -> close -> sync state.

2. Problem Statement

Cloud identity incidents are difficult to manage cleanly when every tool tells only part of the story. A finding may begin in AWS, be discussed in Slack, be tracked in a spreadsheet or table, be investigated in TheHive, and be closed later by an analyst. Without orchestration, the analyst must manually move context between each stage. That creates delay, inconsistency, and gaps in operational visibility.

Why the problem matters in a SOC

- Identity incidents often move quickly and touch multiple teams and systems.
- Cloud findings need enrichment and context before they become truly actionable.
- Prevention, response, and closure are all part of the operating model, not separate concerns.
- Automation improves consistency even before it improves pure speed.

3. Objective

- Design a prototype that covers the full lifecycle of cloud identity security operations.
- Integrate AWS, n8n, Slack, DataTable, and TheHive 5 into one practical automation architecture.
- Showcase the build as both a working prototype and a professional portfolio artifact.

4. Solution Overview

The prototype uses n8n as the central workflow engine. Incoming AWS findings are normalized, enriched, scored, and turned into TheHive alerts through Flow A. Response requests then trigger Flow B, which executes IAM key containment and promotes the related alert to a case. Flow C adds scheduled, preventive IAM hygiene monitoring through the AWS credential report, while Flow D completes the loop by synchronizing final TheHive case state back into Slack and DataTable.

This is valuable because it moves beyond a single-use automation script. The prototype demonstrates operational continuity across multiple stages of the security lifecycle and shows how a SOC-friendly data model can be kept consistent across chat, tracking, and case-management systems.

5. Tools and Technologies Used

Tool / Technology	Why it was used in this workflow
AWS GuardDuty / Security Hub / IAM / CloudTrail	Provide the identity-finding signal, enrichment APIs, credential reports, and containment control surface.
n8n	Orchestrates the four workflows and ties each system together.
Slack	Provides analyst-facing notifications for triage, containment, hygiene digesting, and closure sync.
n8n DataTable	Acts as the durable internal state layer shared by the workflows.
TheHive 5	Provides alerting, case management, promotion, and final closure state used by Flows A, B, C, and D.
AWS CloudShell / curl / test payloads	Support workflow validation and repeatable lab testing during the build process.

6. How the System Works

Stage	What happens
Flow A	Receives and enriches identity findings, scores them, notifies Slack, updates DataTable, and creates TheHive alerts.
Flow B	Executes containment, updates operational state, and promotes the matching TheHive alert into a case.
Flow C	Runs scheduled IAM hygiene checks, creates digest messages, and opens TheHive alerts for hygiene issues.
Flow D	Polls TheHive for recently closed/resolved cases and syncs final state back into Slack and DataTable.

Plain-English view

- In plain English, the prototype acts like a small cloud-identity SOAR system built around n8n.
- One workflow tells the team what happened, another contains the problem, another keeps checking for hygiene issues, and the last one makes sure the final case outcome is written back everywhere that matters.

7. Workflow and Architecture Evidence

The custom architecture view below summarizes the complete operating model and how the workflows interact with AWS, Slack, DataTable, and TheHive 5.

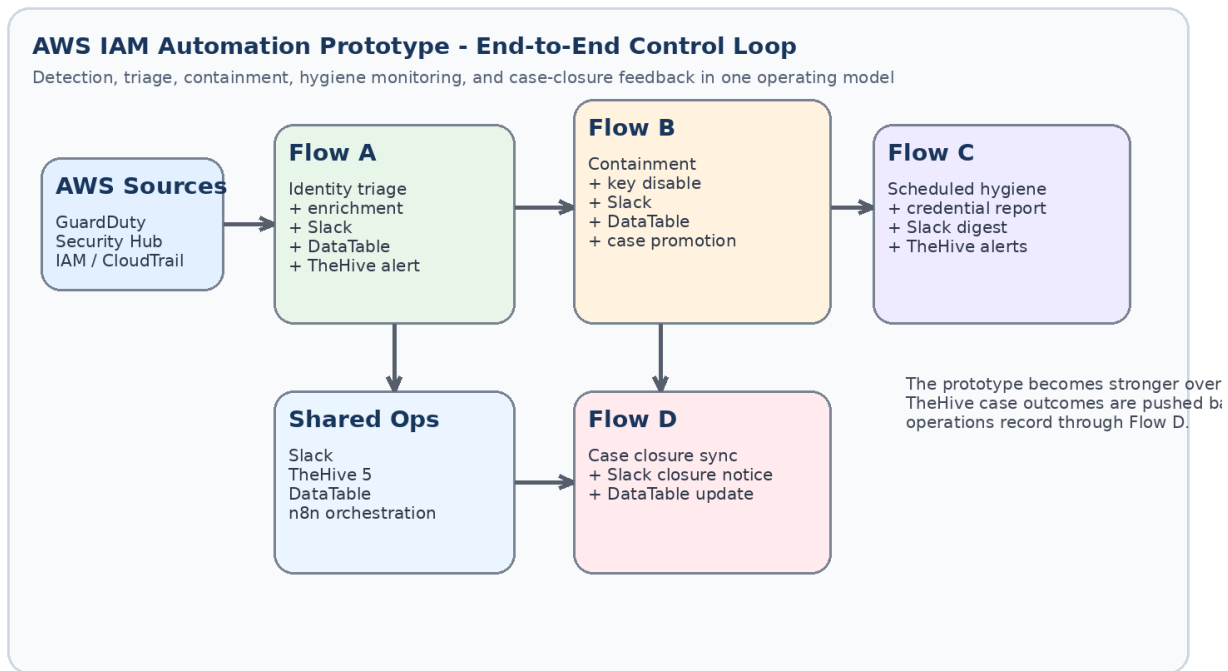


Figure O1. High-level architecture of the four-flow AWS IAM automation prototype.

The master workflow evidence below is split by flow so that each stage of the prototype can be understood independently and as part of the whole.



Figure O2. Flow A - AWS identity triage.



Figure O3. Flow B - AWS identity containment and case promotion.

The preventive monitoring and closure-feedback workflows complete the design by adding proactive hygiene checks and operational state reconciliation.



Figure O4. Flow C - AWS IAM hygiene monitoring.



Figure O5. Flow D - TheHive case closure sync.

8. Key Features and USP

- Four connected workflows covering multiple stages of cloud identity operations.
- Event-driven and scheduled automation in the same prototype.
- Shared DataTable state model across triage, containment, hygiene, and closure.
- Native TheHive 5 integration for alert creation, case promotion, and closure feedback.

- Real validation evidence from n8n, AWS, Slack, DataTable, and TheHive.

9. SOC Value and Expected Impact

The prototype matters because it demonstrates an operating model rather than just a collection of integrations. A SOC can use this kind of design to improve early finding quality, speed up response, reduce analyst context switching, create a durable security-work tracking layer, and keep case-management outcomes aligned with the rest of operations. Even as a prototype, it shows the type of orchestration maturity that makes security workflows easier to run, explain, and extend.

Operational area	Expected effect
Initial understanding	Improved through enrichment and structured triage results.
Response speed	Improved through automated containment and templated case promotion.
Preventive posture	Improved through recurring IAM hygiene checks.
Operational visibility	Improved through Slack and DataTable updates at each lifecycle stage.
Lifecycle completeness	Improved through Flow D closure synchronization back into shared state.

10. Validation and Testing Evidence

The prototype was validated flow-by-flow. Flow A successfully produced TheHive alerts after enrichment and scoring, Flow B promoted those alerts into cases after containment, Flow C created TheHive hygiene alerts from scheduled report findings, and Flow D synchronized case closure back into Slack and DataTable.

 **AWS Identity Finding**

Title: GuardDuty test finding
 Risk: CRITICAL (88)
 Source: guardduty
 Account: 123456789012
 Region: us-east-1
 Principal: lab-identity-test
 Access Key: AKIAVUXCX4MP2BSKC6HI
 IAM Enrichment: iam_last_used_enriched
 IAM Error: none
 Last Used: 1776084180
 Last Service: sts
 Last Region: us-east-1
 CloudTrail Status: cloudtrail_no_events
 CloudTrail Count: 0
 CloudTrail Latest Event: unknown
 CloudTrail Latest Source: unknown
 CloudTrail Latest Time: unknown
 CloudTrail Latest User: unknown
 Finding ID: gd-finding-001
 TheHive Alert Status: thehive_alert_created
 TheHive Alert Ref: guardduty:gd-finding-001
 TheHive Alert ID: ~4636816
 TheHive Alert Error: none

Figure O6. Flow A final Slack output with TheHive alert metadata.

 **AWS Identity Containment**

Action: ContainAccessKey
 Result: containment_executed
 Account: 388096320287
 Region: us-east-1
 Principal: disposable
 Access Key: AKIAVUXCX4MPXCWQGFJ6
 Finding ID: securityhub-test-finding-001
 Summary: Access key AKIAVUXCX4MPXCWQGFJ6 for disposable was set to Inactive.
 Error: none
 TheHive Search: thehive_alert_found
 TheHive Action: thehive_alert_promoted_to_case
 TheHive Comment: thehive_case_comment_created
 TheHive Alert Ref: guardduty:gd-finding-001
 TheHive Alert ID: ~4636816
 TheHive Case ID: ~4436056
 TheHive Alert URL: <http://100.30.129.36:9000/index.html#/alert/~4636816>
 TheHive Case URL: <http://100.30.129.36:9000/index.html#/case/~4436056>
 TheHive Error: none
 Automated with this [n8n workflow](#)

 100.30.129.36
TheHive
 Enterprise Security Incident Response Platform

Figure O7. Flow B final Slack output with promoted TheHive case metadata.

The hygiene and closure feedback validations show that the prototype moved beyond a simple event-handling demo and into a more complete operational model.

 **AWS IAM Hygiene Digest**

Run status: issues_found
 Account: 388096320287
 Region: us-east-1
 Generated: 1776164098
 Users checked: 3
 Missing MFA: 3
 Stale keys: 0
 Unused keys: 0
 Findings: 3
 Report status: credential_report_ready
 Report error: none

Top issues:

1. IAM hygiene: MFA missing for disposable | principal=disposable | risk=HIGH (72)
2. IAM hygiene: MFA missing for lab-identity-test | principal=lab-identity-test | risk=HIGH (72)
3. IAM hygiene: MFA missing for n8n | principal=n8n | risk=HIGH (72)

Automated with this [n8n workflow](#)

Figure O8. Flow C final Slack digest after successful hygiene evaluation.

 **AWS Identity Case Closure Sync**

Flow: flow_d_thehive_case_closure_sync
 Case Title: AWS-ID GuardDuty test finding
 Case ID: ~4436056
 Case Number: 1
 Status: Other
 Assignee: underscore@underscore.co
 Source: n/a
 Source Ref: gd-finding-001
 Principal: lab-identity-test
 Account: 123456789012
 Region: us-east-1
 Updated At: 1776175979396
 Automated with this [n8n workflow](#)

Figure O9. Flow D closure sync message after TheHive case resolution.

The final TheHive views confirm that the platform was used both for alert-level intake and for case-level investigation tracking. This strengthened the prototype significantly compared with a Slack-only or DataTable-only design.

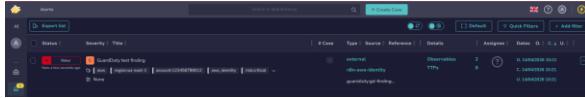


Figure O10. TheHive alert created by Flow A.

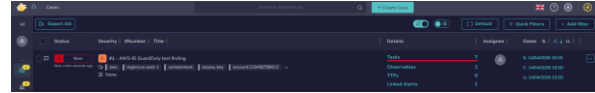


Figure O11. TheHive case created by Flow B and later consumed by Flow D.

11. Real-World Use Cases

- Cloud SOC prototype demonstration for portfolio and interview discussion.
- Identity-focused SOAR proof of concept for AWS environments.
- Education and lab work showing how multiple workflows can cooperate around a shared state model.

12. Skills Demonstrated

- Multi-workflow system design rather than single-flow automation.
- Cloud identity security monitoring and response logic.
- Cross-platform integration across AWS, n8n, Slack, DataTable, and TheHive 5.
- Troubleshooting, iteration, and redesign through validation evidence.
- Professional documentation and portfolio packaging of a working security prototype.

13. Prototype Limitations

- The project is a prototype and was validated in a lab-style environment rather than a live production tenancy.
- Some steps rely on prepared credentials, case templates, and lab-friendly assumptions.
- Quantitative improvements to MTTD or MTTR are expected benefits, not measured production metrics in this document.

14. Future Enhancements

- Refactor shared logic into reusable subworkflows for easier maintenance.
- Add analyst approval gates, dashboards, and richer metrics reporting.
- Expand the identity coverage to multi-account and more complex principal types.
- Add remediation and ticketing patterns beyond the current prototype scope.

15. Conclusion

This project demonstrates a strong, practical prototype for AWS IAM security automation using n8n. It shows not only that separate workflows can be made to work, but that they can be combined into a coherent lifecycle: triage, contain, monitor hygiene, and synchronize closure state. As a portfolio project, it showcases design thinking, cloud security operations understanding, and hands-on multi-tool integration at a meaningful level.